

ShopEZ E-Commerce Shopping Platform

- Technology Stack

Date: 18 June 2026

Team ID: TEAM-2026-001

Project Name: ShopEZ E-Commerce Shopping Platform

Technical Architecture

ShopEZ follows a 3-tier MERN architecture consisting of a React.js frontend, an Express.js and Node.js backend API layer, and a MongoDB database. JWT authentication secures protected routes, while cloud deployment ensures scalability, availability, and performance.

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Responsive web application for customers and administrators	React.js, Bootstrap, React Router
2	Authentication & Authorization	User login, registration, and role-based access control	Node.js, Express.js, JWT, bcrypt.js
3	Product Management	Product catalog, categories, inventory, and CRUD operations	Express.js, Mongoose
4	Shopping & Orders	Cart, checkout, order processing, and tracking	Node.js, Express.js
5	Database	Persistent storage for users, products, orders, and inventory	MongoDB, Mongoose ODM
6	Cloud Database	Managed cloud database service	MongoDB Atlas
7	File Storage	Product images and media assets	Cloudinary / AWS S3
8	Payment Gateway	Secure online payment processing	Razorpay / Stripe
9	Notifications	Order confirmation and status updates	Nodemailer, Email APIs

10	Analytics Dashboard	Sales reports and business analytics	Chart.js, React Charts
11	Infrastructure	Cloud deployment and scalability	AWS / Render / Docker / Nginx

Table 2: Application Characteristics

S.No	Characteristics	Technology / Description
1	Open-Source Frameworks	React.js, Node.js, Express.js, MongoDB, Bootstrap, Chart.js
2	Security Implementations	JWT Authentication, bcrypt hashing, HTTPS, CORS, Helmet, Input Validation
3	Scalable Architecture	Stateless REST APIs, MongoDB Atlas, Docker Containers, Load Balancing
4	Availability	Cloud Deployment, Monitoring, Health Checks, Automated Restart Services
5	Performance	Database Indexing, Caching, Optimized Queries, Lazy Loading, CDN Support